Getting Started with the Pac-Man Game Environment

PREREQUISITES: At this stage you should have (i) a GitHub account; (ii) an account on the evaluation server and; (iii) a copy of the Pac-Man Game environment on your local machine. If you don't satisfy these prerequisites, refer to the Assignment Installation Guide.

In this document we describe the functioning of the Pac-Man game environment, the organisation of its source code and highlight various functions and utilities which you may find useful in your implementation. We'll cover

- The fundamental game loop and game rules (incl termination conditions)
- The **`GameState`** object and its main functions/features
- The organisation of the code
- The purpose and organisation of the layouts

1. The Game Loop

When **`pacman.py`** is run the initial game configuration is loaded from the specified map layout (**`.lay`** file). Controllers are then initialised and any agents mentioned in the layout file are added to the map. The core gameplay loop then begins.

The game is divided up into iterations, where each iteration begins with the game environment asking for an action from the Pac-Man controller. The chosen action is then applied resulting in an update to the **`GameState`** object (more on this in the next section). If there are any ghosts on the map, their controllers are queried next, again resulting in updates to the **`GameState`**. The process continues in a loop as follows:

`GameState -> [Pac-Man action] -> GameState -> [Ghost 1 action] -> GameState-> … -> [Ghost N action] -> GameState -> [Pac-Man action] -> …`
Certain actions from Pac-Man are rewarded by points from the game environment. The point scoring opportunities (which stack) are as follows:

- Eat a food dot: +10
- Eat a "scared" ghost: +200
- Eat the last food dot: +500

Ghosts temporarily enter a "scared" state when Pac-Man eats a power pellet (they look like bigger versions of food dots).

The game ends either when one of the following termination conditions are satisfied:

- Pac-Man has eaten all the food dots on the board
- Pac-Man is caught by a ghost.
- Time has run out.

2. The GameState class

The current state of the game (configuration of the maze, position of the agents, locations of remaining food and the current score) is stored in a **`GameState`** object.

This is an important class that you should become familiar with. Its implementation can be found in the file **`pacman.py`**. When an agent is asked to select an action they will have access to the current state of the game via an instance of this class. Some potentially useful methods/attributes of the **`GameState`** class are the following:.

- **`getLegalActions`** - Given agent index i, returns a list of the legal actions.
- **`generateSuccessor`** - Takes an agent index i and action and creates a new GameState object that would result from agent i taking the given action.
- **`getPacmanPosition`** - Return an (x,y) tuple with the location of Pac-Man on the grid
- **`getGhostStates`** - Returns a list of AgentState (defined in **`game.py`**) objects that store information about a ghosts state. Primarily useful for checking how long ghosts will be scared for when Pac-Man has eaten a capsule.
- **`getScore`** - Returns Pac-Man's current score
- **`getCapsules`** - Returns a list of (x,y) positions of the capsules on the grid
- **`getNumFood`** - Returns the number of food remaining on the board.
- **`getFood`** - returns a 2D array where the value at array position (x, y) is true if there exists a food dot at the corresponding location on the grid.
- **`getWalls`** - returns a 2D array where the value at array position (x, y) is true if the corresponding position is a wall (i.e., not traversable).
- **`hasWall`** - Given an x ,y location, returns **`True/False`** if that location is wall.
- **`hasFood`** - takes an x ,y location as input and returns **`True`** if that location has food and **`False`** otherwise.
- **`isWin`** - **`True`** if the current game state is a win for Pac-Man and **`False`** otherwise.
- **`isLose`** - **`True`** if the current game state is a loss for Pac-Man and **`False`** otherwise.

There are more attributes and methods that may be useful and you should read the code for the GameState class to get the specific details of the methods above. What is listed above is just to get you started. If you are beyond "getting started" then it would also be worth reading the AgentState class in game.py.

3. Agent Controllers

The behaviour of every agent in the Pac-Man game is described by a corresponding controller. All controllers inherit from a base class **`Agent`** (defined in **`game.py`**) which defines a single method, **`getAction`**, that must be implemented.

The simulator interacts with each of the agents in the game via the **`getAction`** method. This method takes a **`GameState`** parameter (see above) and returns a next action for the agent to take, specified via the **`Directions`** (defined in **`game.py`**) class. To see an example agent, take a look at the implementation in **`GoWestAgent.py`**. The different controllers you must develop in this unit will all implement this method differently.

Most of the agents you create will implement **`getAction`** in a way that is reactive to the current game state, that is, they receive a **`GameState`** object, perform some computations and then return an action.

In Part 1 of the Assignment 1 we will work with a specialised type of agent called **`SearchAgent`**, whose implementation can be found in the file **`SearchAgents.py`**. These agents are initialised with a search problem to solve (you'll implement these) and a search algorithm to solve that problem (you'll also implement these).
When a **`SearchAgent`** solves a problem the resulting plan, a sequence of actions, is saved. When the **`getAction`** method of this agent is called, the next action in the plan is returned and executed. The problem is considered solved when the agent has executed all the saved actions.

You should familiarise yourself with the **`SearchAgent`** class. The problem and solver are given in the **`__init__`** method. The plan is computed and saved in the **`registerInitialState`** method. The actions are read from the plan in the **`getAction`** method.

4. Helpful Utilities

The file **`util.py`** provides a number of helpful data structures. You may find it helpful to refer to these implementations when developing your controller. They include:

- Stack
- Queue
- Priority Queue

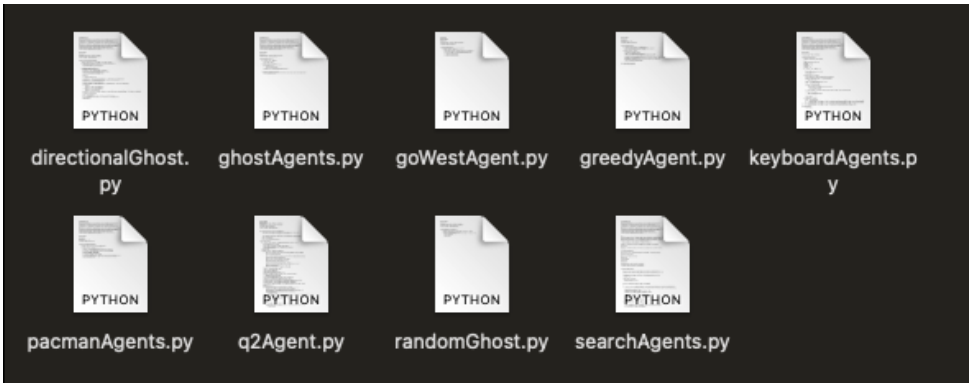
5. How the code is organised

In this section we describe the organisation of the pacman project files. You will need to navigate, read and understand different parts of the project in order to develop your controller. The code is organised into the following directories:

- **`[project root]`**
  - **`|- agents`**
  - **`|- layouts`**
  - **`|- logs`**
  - **`|- problems`**
  - **`|- solvers`**

5.1 Agents

The agents directory contains code for the different agents that participate in the game, so Pac-Man of course and the ghosts. The contents of the directory should look like this.



Important files

- **`SearchAgents.py`**: This file contains the base class implementation for agents that you will develop in Part 1 of the assignment. A description of how a **`SearchAgent`** agent works is given in Section 3. You will not need to modify this file but you should be familiar with how the implementation works.
- **`q2Agent.py`**: You will modify this file in Part 2 of the assignment. You will implement *Alpha-Beta* search in the **`getAction`** method of this agent.

5.2 Layouts

The layouts directory contains maps that you can use to test your implementations for each question. **These are not the same maps that we evaluate you on, but they are similar.** The prefix of the filename should reflect the question it should be used in. smallClassic.lay and testMaze.lay are also included as part of the demo for this document. An example of the contents of a layout file is shown below.

```
%%%%%%%%%%%%%%%%%%%%
%......%G  G%......%
%.%%...%%  %%...%%.%
%.%o.%........%.o%.%
%.%%.%.%%%%%%.%.%%.%
%........P.........%
%%%%%%%%%%%%%%%%%%%%
```

Explanation of symbols

- `%` = Wall (non-traversable location)

- **`P`** = The starting location of Pac-Man
- **`G`** = The starting location of a ghost
- **`.`** = A food dot
- **`o`** = A power pellet / capsule

5.3 Logs

The **`logs`** directory stores the log files that are created each time you run pacman.py with the `-o` flag. More information about the logs can be found in Section 8.

5.4 Problems

This directory contains a python file for each of the three tasks in question 1 of the assignment:

- **`q1a_problem.py`**:
- **`q1b_problem.py`**:
- **`q1c_problem.py`**:

All of these files contains implement the same three methods:

- **`getStartState`**
- **`isGoalState`**
- **`getSuccessors`**

You will implement each class to describe the problem representation for each of the questions in Part 1 of the assignment. More information about how to approach this task (and Part 2!) can be found in Section 6.

5.5 Solvers

There are three files in this directory. Each one contains a function that stub where you will implement the corresponding search strategy for each of the questions in Part 1 of the assignment.

- **`q1a_solver.py`**, which contains the **`q1a_solver`** function stub.
- **`q1b_solver.py`**, which contains the **`q1b_solver`** function stub.
- **`q1c_solver.py`**, which contains the **`q1c_solver`** function stub.

6. Implementing your controller

In Sections 6.1 and 6.2 we give an overview of how you might approach Q1(a). This information is also useful as a general guide for Q1(b) and Q1(c). Then, in Section 6.3, we give an overview of how you might approach Q2.

6.1 Model the problem (Assignment Part 1)

The first thing to do is model the problem concretely, which you will achieve by modifying the classes **`q1a_problem`**, **`q1b_problem`**, and **`q1c_problem`**. Constructed by the game environment, this class stores the initial state of the game board (a **`GameState`** object) and a series of function stubs, which you will implement. The stubs are as follows:

- **`getStartState`:** this function takes no parameters and returns the **initial state** of the problem (this is a data structure analogous to but not necessarily the same as GameState); i.e., it represents the configuration of the map at the start of the game.
- **`isGoalState`:** given a **current** state, this function returns true if that state satisfies the goal conditions of the problem (i.e., you get to decide what those are).
- **`getSuccessors`:** given a **current** state, this function returns a set of states, each reachable from the current state by applying a valid action having an associated cost.

As part of this task, you will get to decide which features of the environment (== GameState) need to be represented in your corresponding search state. **You get to decide what the search state data structure looks like**, and which attributes it contains (e.g., the tentative position of Pac-Man seems like it might be a helpful thing to store!)

6.2 Implement a search algorithm (Assignment Part 1)

With a model of the problem in hand, you must turn your attention to the next issue: how to compute a solution. Here you are asked to implement the function stub **q1a_solver** which is found in the **solvers** directory**.**

The function **`q1a_solver`** is called by the search agent when it is initialised. It receives an instance of the problem model (that you wrote earlier) as a parameter. Its job is to return a sequence of actions which moves Pac-Man from start to target. Implement this stub by undertaking a state-space search. Refer to the functions in your problem model to initialise the search (**`getStartState`**), to advance the search (**`getSuccessors`**) and test for a solution (**`isGoalState`**). **Look for the key data structures in util.py to make implementing your solution easier.**

The expected output from the search function is a list of actions denoted using the **`Directions`** class in **`game.py`**, that fully specifies the plan for Pac-Man. When the search returns this plan the simulator executes every action of the plan and scores the performance of Pac-Man. If the actions in the plan are exhausted (i.e., all have been executed) then the game will end. The game will also end when Pac-Man has collected all the food dots. In Q1(a) we expect the game to end when the single food dot is collected. However in Q1(b) we expect you to only collect a single food dot. In Q1(c) the game may end either way, depending on whether your algorithm decides that collecting every food dot is worth it. In any case…

Make sure you solve the problem completely before returning from this function!

6.3 How to Approach Part 2 (of the Assignment)

For this part of the assignment we will use the full **`GameState`** data structure from the Pac-Man environment -- instead of writing our own custom state-space representation as in Part 1. The **`GameState`** data structure contains a range of (now useful) features that you may want to consider when developing your controller (see Section 2).

For the same reason we also no longer need to specify a problem model. Instead, we will use the (implicit) model provided by the game environment. In other words:

- The game environment determines the initial state
  (i.e., we no longer need to write **`getStartState`**)
- The game environment determines when the game is won or lost
  (i.e., we no longer need to implement **`isGoalState`**)
- The game environment provides an implementation of **`getSuccessors`**
  (i.e., we no longer need to implement that, either)

You need to implement your adversarial search in the **getAction** method of the **`Q2_Agent`** class. This method takes a **`GameState`** object representing the current state of the game and should return an action for Pac-Man. You will encounter between 1 and 4 ghosts in this part, but your adversarial search should work for any number of ghosts. Review Section 2 for methods of **`GameState`** that you can use to interact with the ghosts in the game.

Your core algorithm must be alpha-beta search but you are free to implement any optimisations you like. There's no restriction on the depth of the search. Just remember that each game has a 30 second time limit on the server so a deeper search might not necessarily be better.

7. Testing and Debugging

This section discusses how to evaluate your agents and some general strategies to help you when your agents do not behave as you intended.

7.1 Evaluator.py

We've included a python script **`evaluator.py`** that you can use to produce a summary of your performance across all parts of both questions on the sample layouts. To run this script you will need to install **pandas**, **tqdm,** and **tabulate**. With your conda environment active, run the following:

```
> conda install pandas
> conda install tqdm
> conda install tabulate
```

Navigate to the directory with the **`evaluator.py`** script, then run the following.

```
> python evaluator.py
```

Confirm whether or not you have the latest version of the environment code (if you don't stay up to date then you may not be evaluated correctly). The evaluator will attempt to run your implementation for each question on all the sample layouts provided and produce a table summarising your results. Obviously if you have not attempted a question then the evaluator will fail on that question. If you don't want the evaluator to run a particular question then you can add the following:

`--q1a`
: Tells the evaluator to not run q1a layouts

`--q1b`
: Tells the evaluator to not run q1b layouts

`--q1c`
: Tells the evaluator to not run q1c layouts

`--q2`
: Tells the evaluator to not run q2 layouts

For example, the following will only run q1c and q2.

```
> python evaluator.py --q1a --q1b
```

The scores you see here are not the same as your scores on the evaluation server because these layouts are different. This script has been included to give you a holistic view of your performance, and to aid you in performing experiments.

7.2 Debugging

This section contains some general tips and strategies for how to test and debug your agent controllers. These strategies will help you to identify how and when a problem occurs, and they may be helpful to form a hypothesis about what types of problems could be occurring in your implementation. Try these strategies **before you ask for help**.

- **Change parameters** - You might find your agent is performing well on some layouts and not others. Try a different layout and see where your agent stops performing the way you expect it to. You might find there is a specific piece of the layout that is causing your agent trouble. Or it may be that the inclusion of capsules or more ghosts is problematic. Watch your agent in a variety of scenarios and you will likely isolate the problem.
- **Inspect the log files** - The log files should contain timestamps with some function calls, observations that your agent receives, actions that you agent takes, states, and other metrics. Look for anything unexpected. Combine this with the first tip to connect what you're observing in Pac-Man's behaviour with the way Pac-Man is interacting with the simulator based on your code.

- **Run multiple games** - You can add -n as a command line argument to run multiple games on the same layout (see Section 9 for details). This is likely to be more useful in part 2 of the assignment than part 1.
- **Use a text output** - You can view the game as a textual output by including -t as a command line argument (see Section 9 for details). Unlike the graphical output, the textual output allows you to view the entire history of the game and it doesn't disappear when the game finishes.
- **Master the small layouts before tackling the big ones** - There are some very small layouts and some very large layouts. If your agent hasn't mastered the small ones then it is unlikely that testing them in the large layouts will help. If your agent is performing very well but there are some specific edge case type scenarios where it doesn't perform as expected, then it's more likely you'll encounter these scenarios on a smaller map than a larger one.
- Use the **-c** flag! Without this flag the game environment will not catch and process timeouts and exceptions
- **Use print statements in your code** - A single well placed print statement can be all it takes to debug your code. They're easy to use and incredibly useful.
- **Use the python debugger** - A little more work than print statements but it may be more useful.

What if none of these steps work?
Check Ed to see if someone else has had a similar problem to you. You can't post specifics about your solution to the assignment, but you may find something useful there. If there is a problem with our code and not yours then chances are someone else has already encountered it. If not then you can make a post and maybe help everyone out.

Where else can I get help?
The best thing is to come to consultation sessions or speak with your tutor in class.

When should you seek help?
As early as possible! The teaching team cannot help you directly implement your controller but they can help you improve your understanding of search algorithms, and they may be able to point you in the right direction if you are having trouble understanding the operation of the game environment.

It may happen that a bug in your code is the result of code that we (the teaching team) provided. If you suspect that a bug is not in your code then you should check (and post) on Ed. It's a good idea to check these pages regularly even if things are going well.

What if the bug is in the game code provided?
Check out Section 10 for details on how to report a suspected issue in our code.

8. Log files

When you run the code with the `-o` flag, the logger is instantiated and logs function names, layout, number of agents, observation in each loop, action taken in each loop, state after action is taken, move history of PacMan, and other metrics with time stamps for you.

Log files are saved in the **`logs`** directory. You can use the log files to check the state of the PacMan board during the game, the actions PacMan has taken, and the final results. All metrics logged are named. If you are unsure, what exactly is logged, look up what function name was logged before the metric and have a look into the code of that function.

9. Command Line Parameters

You can instantiate the **`pacman.py`** script with different parameters, which affect the type of game that will be played. You should become familiar with these options, as they will be helpful during your development and testing!

```
Usage:
       USAGE:       python pacman.py <options>
       EXAMPLES:   (1) python pacman.py
                    - starts an interactive game
                  (2) python pacman.py --layout smallClassic --zoom 2
                  OR  python pacman.py -l smallClassic -z 2
                        - starts an interactive game on a smaller board, zoomed in


Options:
  -h, --help               show this help message and exit
  -n GAMES, --numGames=GAMES
                           the number of GAMES to play [Default: 1]
  -l LAYOUT_FILE, --layout=LAYOUT_FILE
                           the LAYOUT_FILE from which to load the map layout
                           [Default: mediumClassic]
  -p TYPE, --pacman=TYPE
                           the agent TYPE in the pacmanAgents module to use
                           [Default: KeyboardAgent]
  -t, --textGraphics       Display output as text only
  -q, --quietTextGraphics
                           Generate minimal output and no graphics
  -g TYPE, --ghosts=TYPE
                           the ghost agent TYPE in the ghostAgents module to use
                           [Default: RandomGhost]
  -k NUMGHOSTS, --numghosts=NUMGHOSTS
                           The maximum number of ghosts to use [Default: 4]
  -z ZOOM, --zoom=ZOOM  Zoom the size of the graphics window [Default: 1.0]
  -f, --fixRandomSeed   Fixes the random seed to always play the same game
  -r, --recordActions   Writes game histories to a file (named by the time
                           they were played)
  --replay=GAMETOREPLAY
                           A recorded game file (pickle) to replay
  -a AGENTARGS, --agentArgs=AGENTARGS
                           Comma separated values sent to agent. e.g.
                           "opt1=val1,opt2,opt3=val3"
```

```
-x NUMTRAINING, --numTraining=NUMTRAINING
                        How many episodes are training (suppresses output)
                        [Default: 0]
--frameTime=FRAMETIME
                        Time to delay between frames; <0 means keyboard
                        [Default: 0.1]
-c, --catchExceptions
                        Turns off exception handling and timeouts during games
--timeout=TIMEOUT       Maximum length of time an agent can spend computing in
                        a single game [Default: 30]
-o, --outfile           output file name
```

10. Reporting Bugs

If you happen to stumble upon a bug in our code please:

1. Check to see if someone else has posted the issue already.
2. Make a post on Ed and include all the information to reproduce the issue.

11. Additional resources

Original Pacman page from Berkely pacman: http://ai.berkeley.edu/project_overview.html

What to do if you find source code online

- **You can be inspired but you need to acknowledge the source, discuss the approach in your report and then talk about your extensions/modifications**

If you use generative AI tools, i.e. ChatGPT, you must acknowledge their use and supply the prompts used with your submission.

Don't cheat, serious penalties apply!